

C++ BASIC PROGRAMMING ASSIGNMENT

Problem Statements: if-else, switch-case and Loops

Instructions: Write a separate C++ program for each problem. Take required values from the user, display clear output, and use the user defined functions.

Part A – if-else Statements

1. Positive, Negative or Zero

Write a program to accept an integer and determine whether it is positive, negative, or zero using if-else.

2. Even or Odd

Write a program to accept an integer and check whether the number is even or odd.

3. Largest of Two Numbers

Write a program to accept two integers and display the larger number. Also handle the case when both numbers are equal.

4. Largest of Three Numbers

Write a program to accept three numbers and display the largest number using if-else statements.

5. Grade Calculator

Accept marks obtained by a student and display the grade using the following criteria: 90–100: A, 75–89: B, 60–74: C, 40–59: D, below 40: Fail. Validate marks outside 0–100.

6. Leap Year

Write a program to accept a year and determine whether it is a leap year.

7. Simple Electricity Bill

Accept the number of electricity units consumed and calculate the bill using slab rates of your choice. Display the total bill and explain the slabs used.

Part B – switch-case Statements

8. Simple Calculator

Create a menu-driven calculator using switch-case. The user should select addition, subtraction, multiplication, division, or modulus and enter two numbers.

9. Month Name

Accept a number from 1 to 12 and display the corresponding month name using switch-case.

10. Vowel or Consonant

Accept a character and use switch-case to determine whether it is a vowel. If it is not a vowel, display that it is a consonant.

11. Menu-Driven Area Calculator

Create a menu using switch-case to calculate the area of a circle, rectangle, square, or triangle based on the user's choice.

12. ATM Menu

Create a simple ATM menu using switch-case with options such as Check Balance, Deposit, Withdraw, and Exit. Maintain the balance during the program execution.

Part C – Loops (for, while and do-while)

13. Print Numbers

Write programs using for, while, and do-while loops to print numbers from 1 to N.

14. Sum of Natural Numbers

Accept N and calculate the sum of the first N natural numbers using a loop.

15. Multiplication Table

Accept a number and print its multiplication table from 1 to 10 using a loop.

16. Factorial

Accept a positive integer and calculate its factorial using a loop.

17. Reverse a Number

Accept an integer and reverse its digits using a while loop.

18. Palindrome Number

Accept an integer and determine whether it is a palindrome number.

19. Count Digits

Accept an integer and count the number of digits using a loop.

20. Sum of Digits

Accept an integer and calculate the sum of all its digits.

21. Prime Number

Accept an integer and determine whether it is a prime number using a loop.